

Roll No.

TRA-707

B. TECH. (ME-RA)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

AUTOMATIC CONTROL

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define the following terms used in control systems : (CO1)

(i) Controlled and Manipulated variables

(ii) Disturbances

(iii) Feedback control

(iv) Transfer function

(v) Characteristic equation

OR

(b) What do you understand by the open-loop and closed-loop control systems ? Discuss with examples and their respective advantages. (CO1)

P. T. O.

2. (a) What are the poles of a control system ? Describe the following system responses : undamped, overdamped, underdamped, and critically damped systems in terms of the poles mapping on s-plane.

(CO1 & CO3)

OR

- (b) Plot and write the expression (with Laplace transform) for the following signals : step, ramp, sinusoidal, and impulse.
3. (a) Find the solution $x(t)$ of the following differential equation :

(CO1 & CO2)

$$\ddot{x} + 2\dot{x} + 5x = 3; \text{ where } x(0) = 0 \text{ and } \dot{x}(0) = 0.$$

OR

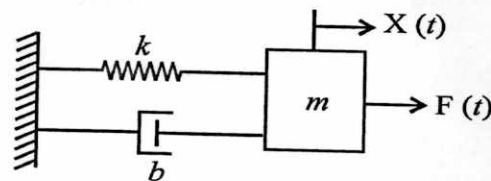
- (b) Find the solution $x(t)$ of the following differential equation :

(CO)

$$\ddot{x} + 3\dot{x} + 2x = 0; \text{ where } x(0) = P \text{ and } \dot{x}(0) = Q.$$

4. (a) Derive the transfer function of the mass-spring-dashpot system as shown in the figure below. Also, comment about the stability and the behaviour of the system if, $m = 1$, $b = 3$, and $k = 5$.

(CO3)



OR

- (b) What do you understand by the Routh's stability criterion ? Comment about the stability of a system, whose characteristic equation is given by the following polynomial :

(CO3)

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

(3)

5. (a) Derive the expression for the unit-step response of a first-order system. Also, plot the response and discuss about the time constant. (CO2)

OR

- (b) Describe the following transient-response characteristics of a control system with a neat sketch : (CO2)

- (i) Delay time
- (ii) Rise time
- (iii) Peak time
- (iv) Maximum overshoot
- (v) Settling time